Jeevan Thapa

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LinkedIn: thapa-jeevan **Website:** thapajeevan.com.np **Scholar:** Google Scholar

Ph.D. candidate specializing in continual learning, parameter-efficient adaptation, and user modeling. First author of papers at ICML and ECIR, with industry research experience developing continual user-modeling methods for recommender systems at Zillow.

RESEARCH FOCUS

Continual Learning and Generalized Category Discovery | Computer Vision | Recommender Systems and User Modeling | Parameter-Efficient Adaptation | Gene-Disease Association

EDUCATION

Ph.D. in Computing and Information Sciences *Aug 2022 – Feb 2028 (Expected)*
Rochester Institute of Technology (RIT) Rochester, NY
GPA: 3.97/4.00
Relevant Coursework: Deep Learning, Statistical Machine Learning, Quantitative Foundations, Non-Convex Optimization for Modern Machine Learning

Bachelor's Degree in Computer Engineering *Nov 2015 – Sep 2019*
Pulchowk Campus, Tribhuvan University Lalitpur, Nepal
Relevant Coursework: Data Mining, Artificial Intelligence, Big Data Analytics, Probability and Statistics

PUBLICATIONS

J. Thapa, R. Li. *Bayesian Adaptation of Network Depth and Width for Continual Learning*. In Proceedings of the 41st International Conference on Machine Learning (**ICML**), 2024.

J. Thapa, S. Zhao, K. Shindo. *Evolving Mixture of Low-Rank Experts for Continual User Modeling*. In Proceedings of the European Conference on Information Retrieval (**ECIR**), 2026.

J. Thapa, R. Li. *Cross-Task Representation Alignment for Exemplar-Free Class-Incremental Learning*. *Under review*.

RESEARCH AND INDUSTRY EXPERIENCE

Graduate Research Assistant *Aug 2022 – Present*
Rochester Institute of Technology Rochester, NY

- Developed a Bayesian continual learning framework that dynamically adapts neural network depth and width for sequential visual recognition tasks; published at **ICML 2024**.
- Developed a cross-task representation alignment framework for exemplar-free class-incremental image classification, improving average accuracy by 2.16 percentage points; manuscript under review.
- Designed a parameter-efficient method for adapting Vision Transformers to continual generalized category discovery in medical imaging using full-covariance Gaussian prototypes.
- Developed a probabilistic graph propagation framework with module diversity regularization that integrates bioinformatics data for gene-disease association prediction; manuscript in preparation.

Research Scientist Intern *May 2025 – Sep 2025*
Zillow Group Remote

- Developed a mixture-of-experts (MoE) architecture with rank-1, LoRA-style experts for continual user modeling, enabling parameter-efficient adaptation across sequential recommendation and user-attribute prediction tasks; published at **ECIR 2026**.
- Designed a semantic expert-selection strategy to reuse transferable knowledge and reduce cross-task interference, outperforming the strongest baselines by 2.88% relative HitRate@5 in recommendation and 1.37% in accuracy for attribute prediction.

Machine Learning Engineer *Sep 2019 – Jun 2022*
Fusemachines Kathmandu, Nepal

- Led the development of a multimodal machine-learning pipeline for identifying potential trafficking activity in online advertisements using video, image, and text data.
- Built image–text contrastive models for advertisement matching, face-image matching for identity linking, and BERT-based social-handle extraction, improving cross-ad linkage accuracy by 35%.
- Developed and deployed a lightweight object detection model on NVIDIA Jetson Nano for real-time visual classification, improving inference throughput by 47%.
- Authored computer vision instructional materials covering deep learning for visual recognition, object detection and image segmentation as part of Fusemachines AI Education Program.

SKILLS

Programming	Python, C++, C
ML Frameworks	PyTorch, Torchvision, scikit-learn, Transformers, PEFT, Transformers4Rec
Data and Experimentation	NumPy, Pandas, NVTabular, MLflow, Matplotlib, Seaborn
Tools and Platforms	AWS, Git, Linux, Bash, Jira, LaTeX

AWARDS AND HONORS

- **RIT Ph.D. Scholarship** - Full financial support through NSF- and NIH-funded research projects.
- **Tribhuvan University Full Scholarship** - Granted by the Government of Nepal after ranking 11th nationwide in the entrance examination (4% acceptance rate).
- **Fusemachines AI Fellowship** - Awarded a merit-based scholarship through a nationwide selection process to pursue the AI MicroMasters program.

TALKS

- **Poster Presentation** - “Bayesian Adaptation of Network Depth and Width for Continual Learning” at **ICML 2024**, Vienna, Austria.
- **Guest Lecture** - “Continual Test-time Adaptation” and “Deep Learning Toolkit” in graduate course CISC-865: Deep Learning at RIT.

ACADEMIC SERVICES

- Conference reviewer for ICLR 2025-2026, ECCV 2026, NeurIPS 2026, AAAI 2026.